

Arduino ~~~~~> HC-SR04

5V ————— VCC

Any digital pin { D2 ————— Echo
D3 ————— Trig

GND ————— GND

Generally Arduino code has 3 section

1) Definition section
 ↳ Define variables

2) Set up () { };

3) loop () { };

```
Const int trigPin = 3;
```

```
Const int echoPin = 2;
```

```
long duration; // duration of pulse travelling
```

```
long distance; // to store the value of  
Distance
```

```
Void setup() {
```

```
Serial.begin(9600); // Baud Rate
```

```
pinMode(trigPin, OUTPUT);
```

```
pinMode(echoPin, INPUT);
```

```
}
```

the Trigger pin and echo pin we define previously, define them that they are for input or output

Here, TrigPin → for output
echo Pin → for Input

```
void loop () {
```

```
digitalWrite (trigPin, LOW);  
delayMicroseconds (2);  
digitalWrite (trigPin, HIGH);  
delayMicroseconds (10);  
digitalWrite (trigPin, LOW);
```

trigPin off
for 2 μ s

on for
10 μ s

off

```
duration = pulseIn (echoPin, HIGH);
```

```
distance = duration * 0.034/2;
```

speed of
ultra sonic
wave

```
Serial.print ("Distance : ");  
Serial.print (distance);  
Serial.println (" cm");
```

// prints String
"Distance" of
LCD
// prints
distance value
from sensor.

```
delay (10);
```

```
}
```

* Pulse In (~~~~, ~~~~);

→ Calculate time (how much time it takes
after strike back from
object)